Transforming a raster image into vector crosshatching

To achieve this, three scripts are successively used:

1. The first one, ***MeF_Doc_Crossed.jsx***, allows you to format the document *(see page 2 for usage)*.

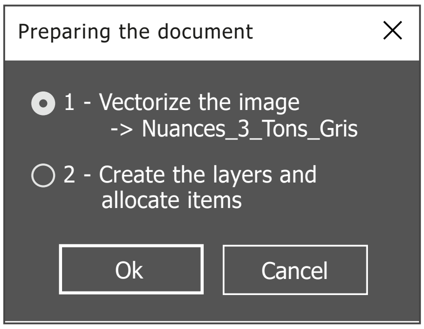
2. The second script, ***supprPetitsObjets.jsx***, is optional. Depending on the size of the original image and the desired result, it allows you to remove paths that are too small. The goal is to save time in the next step, where items too small to create hatching will be removed anyway.

   For the example image used on the following pages, the total processing time was 8 minutes and 19 seconds without using this script, compared to 4 minutes and 17 seconds including the time spent using this script to delete objects whose width **or** height were less than 1 mm. *(see the dedicated pdf for usage)*.

3. Finally, the third, ***crossHatching.jsx***, allows the remaining flat areas to be transformed into vector hatches oriented according to the layer on which they are located *(see page 3 for usage)*.

Prerequisite: save the file [Nuances_3_Tons_Gris.ai] to your computer and update the link to this file on line 30 of this script.

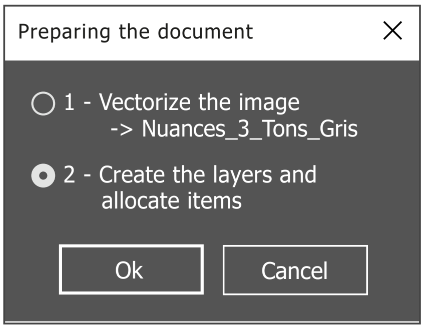
1. Select an RGB image and then run the script. The following dialog box appears:

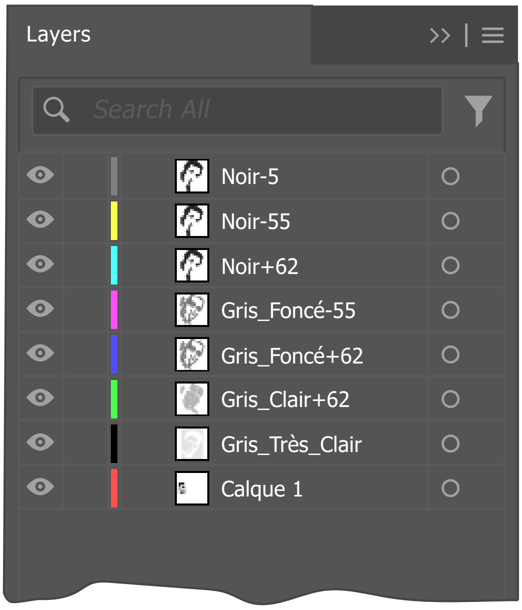


With the first step selected by default, click [OK].

Then, in the "*Image Trace*" palette and in the "*palette*" field, select "Nuances_3_Tons_Gris". Adjust if necessary the slider "Noise". This is also the time to adjust the "*noise*" slider. By using the preview, you can avoid creationg some small paths that add nothing to the final render but would onlu increase processing tme.

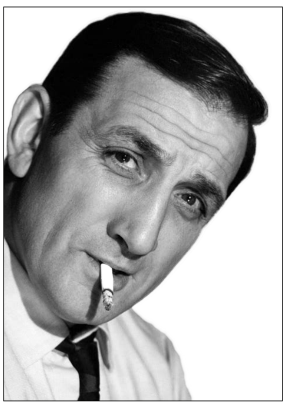
2. Relaunch the MeF_Doc_Crossed.jsx script, select the second step and click "OK".





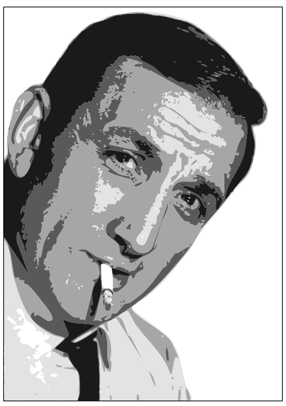
This adds seven new layers to the document with predetermined names, and then distributes the paths across the different layers based on their color.

At the end of the process, the "Gris_Très_Clair" and "calque 1" layers are deleted.



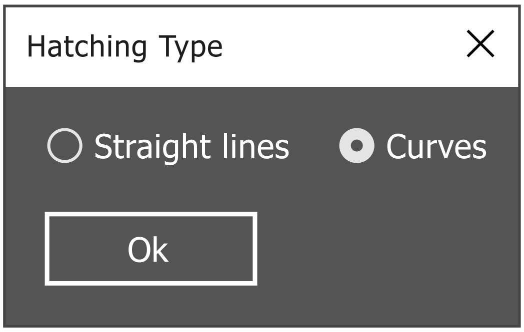
Original image

for information:
474 x 672 px,
72 dpi



Vectorized image using this script.

This script transforms the solid areas of a vector image created with the "10_MeF_Doc_Crossed.jsx" script into lines. This is useful, for example, for reproducing them with a pen mounted on a cutting plotter.



Choosse between straight hatching or curved hatching (as in the example below).

The angle and density of the lines are adjusted according to the layers on which the solid areas to be replaced are located. You may want to proceed in stages (generally by layer) to limit the number of paths to be processed simultaneously.

Depending on the size and number of selected objects, processing time per object ranges from 0.5 to 1 second.

Résult: